



fig-redraw-093 (proposed)

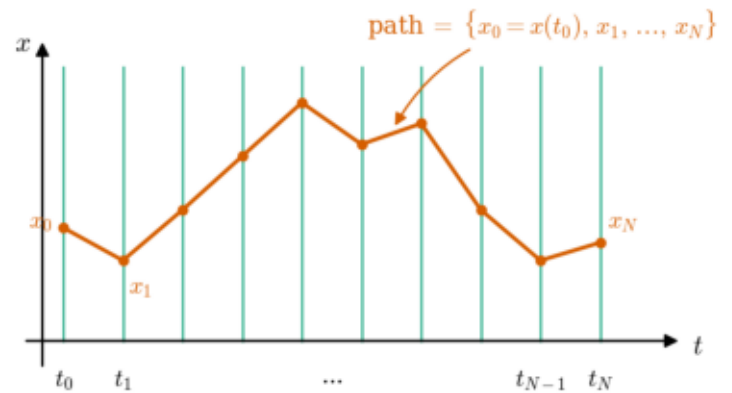


fig-notes-001.jpg (original)
 chapters/ch03-scalar-fields/scalar.tex:358

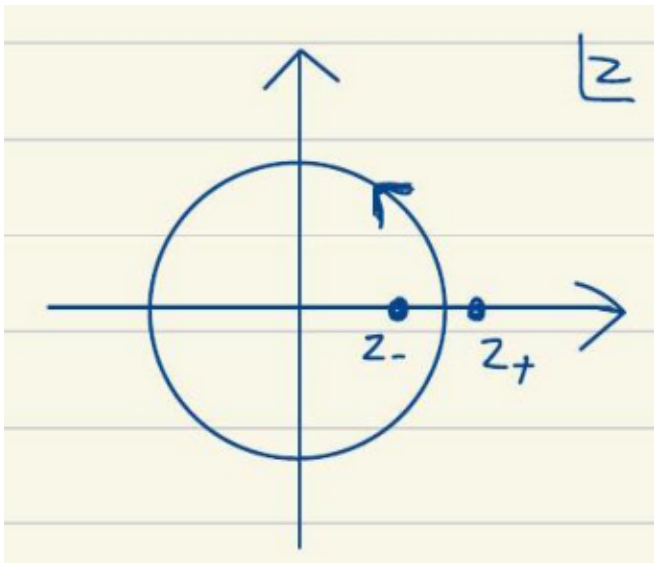


fig-redraw-001 (proposed)

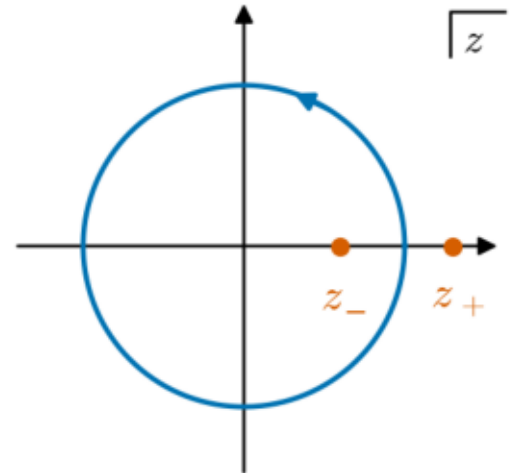


fig-notes-002.jpg (original)
 chapters/ch03-scalar-fields/scalar.tex:383

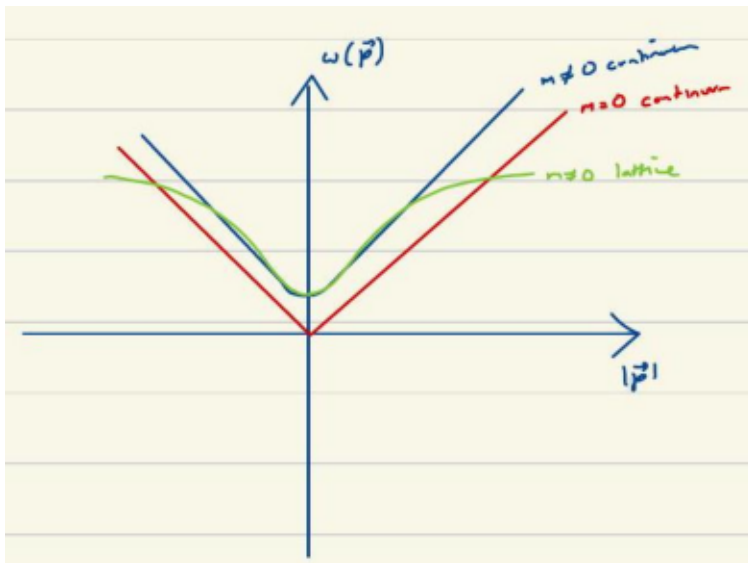


fig-redraw-002 (proposed)

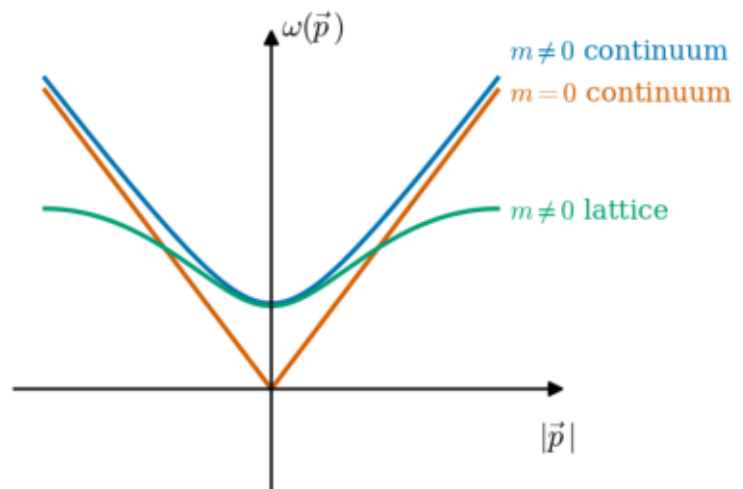


fig-notes-061.png (original)
chapters/ch04-gauge-fields/gauge.tex:117

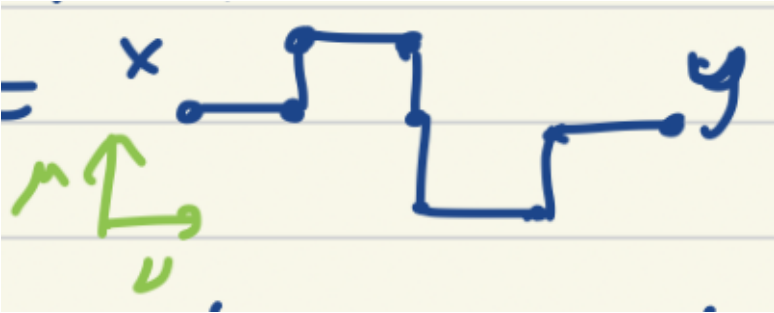


fig-redraw-061 (proposed)



fig-notes-064.png (original)
chapters/ch04-gauge-fields/gauge.tex:715

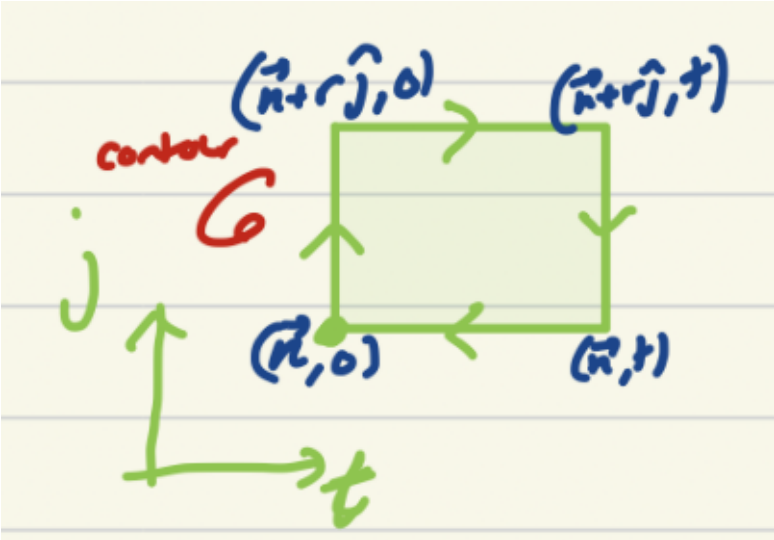


fig-redraw-064 (proposed)

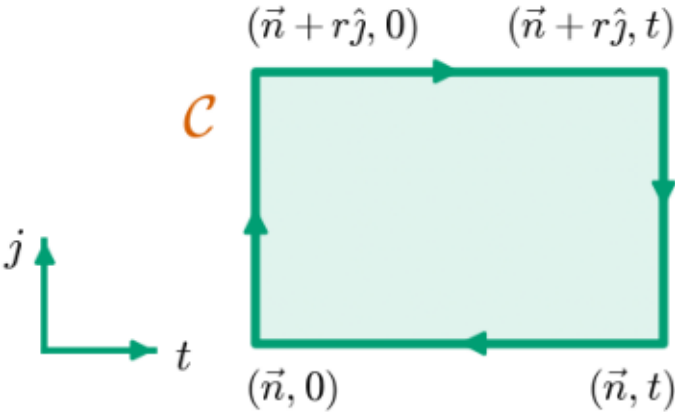


fig-notes-016.jpg (original)
chapters/ch04-gauge-fields/gauge.tex:748

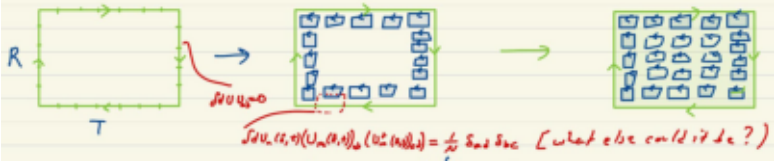


fig-redraw-016 (proposed)

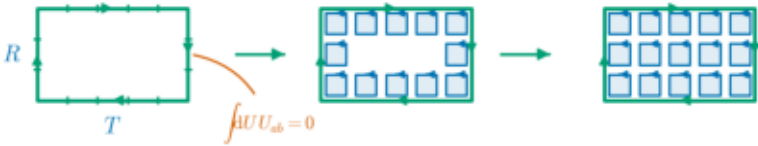


fig-notes-058.png (original)
chapters/ch04-gauge-fields/gauge.tex:762

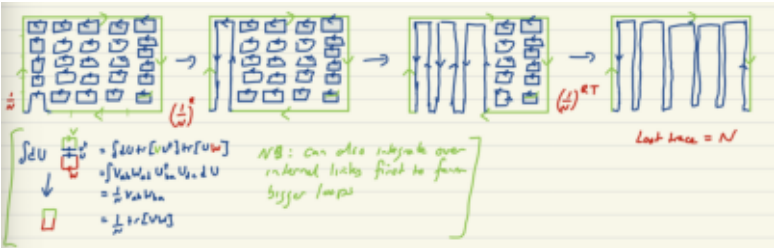


fig-redraw-058 (proposed)

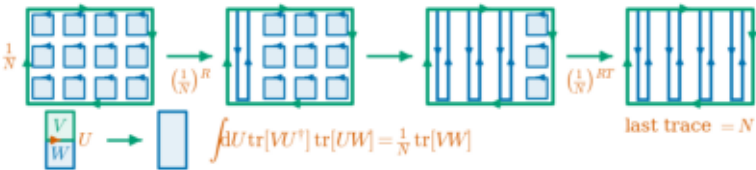


fig-notes-066.png (original)
chapters/ch04-gauge-fields/gauge.tex:884

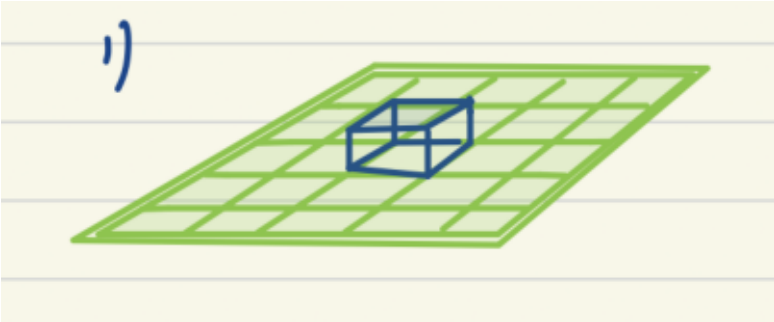


fig-redraw-066 (proposed)

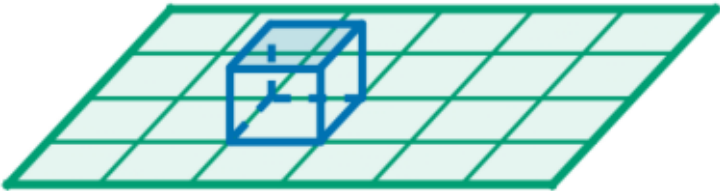


fig-notes-067.png (original)
chapters/ch04-gauge-fields/gauge.tex:886

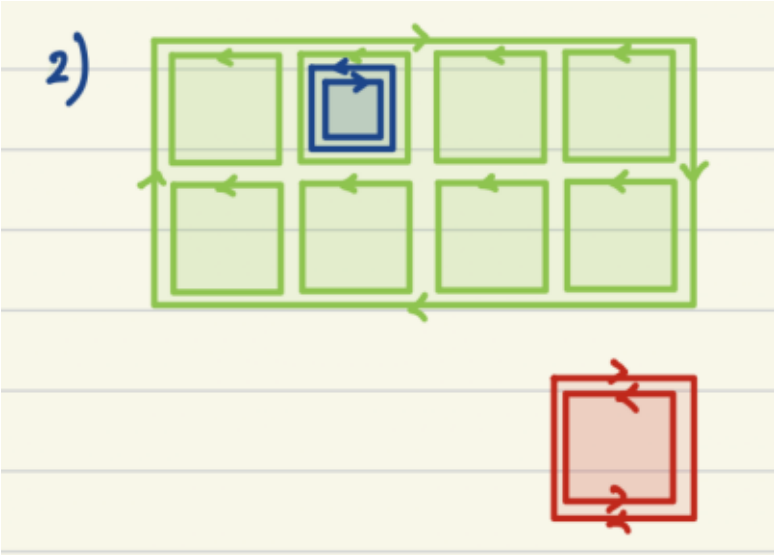
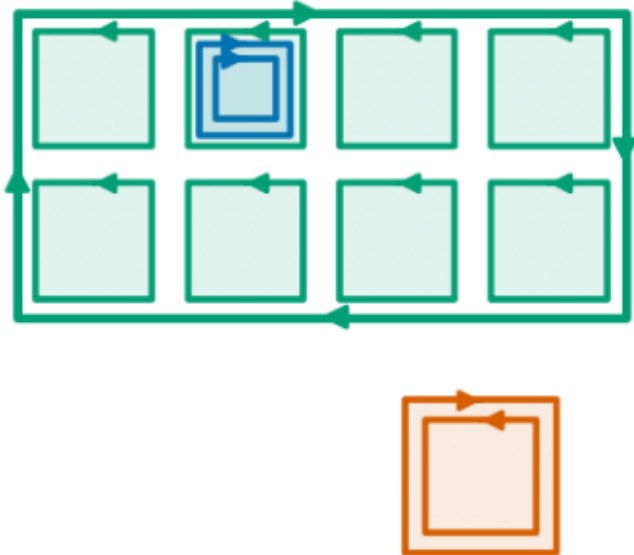


fig-redraw-067 (proposed)



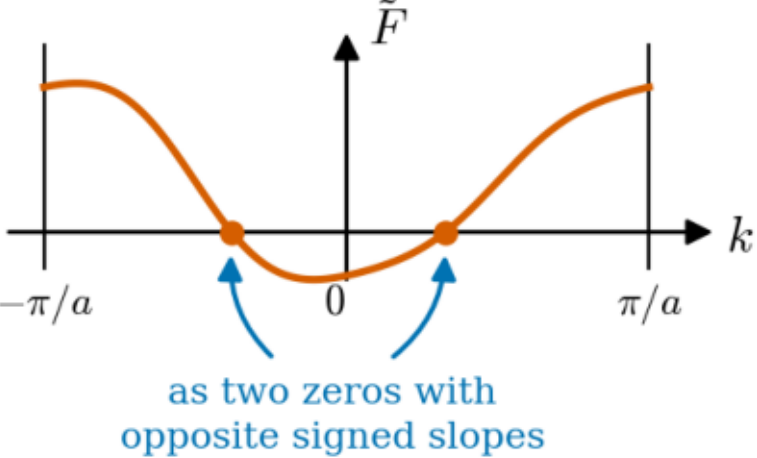
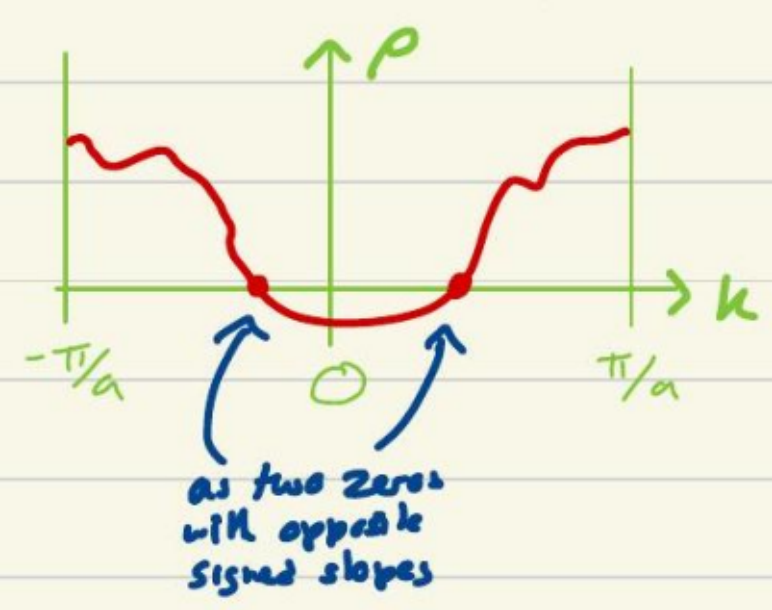
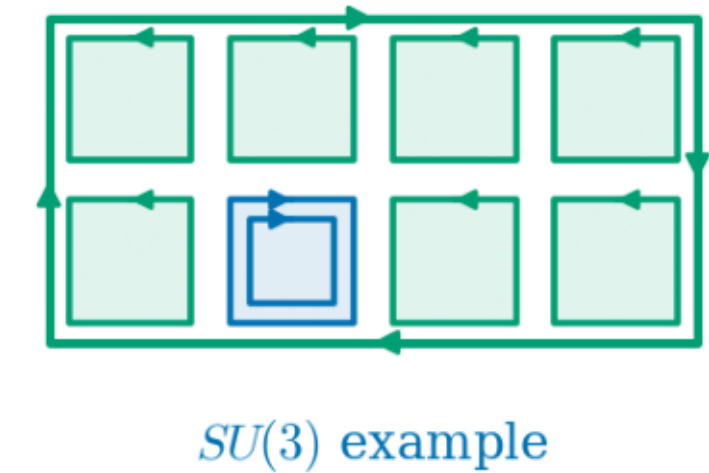
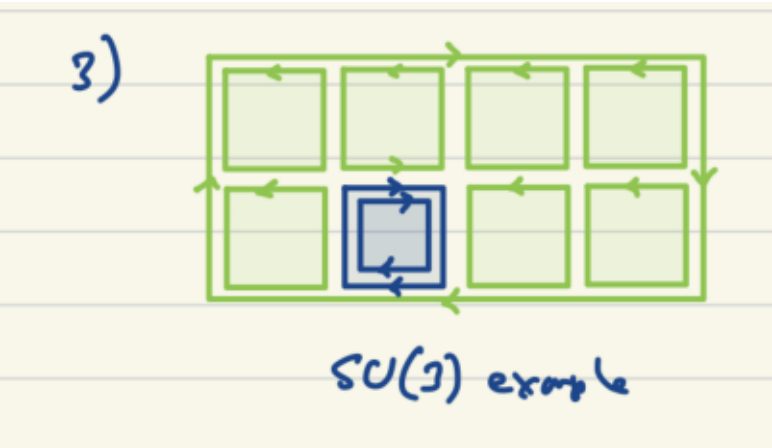


fig-notes-017.jpg (original)
 chapters/ch06-lattice-fermions/fermions.tex:327

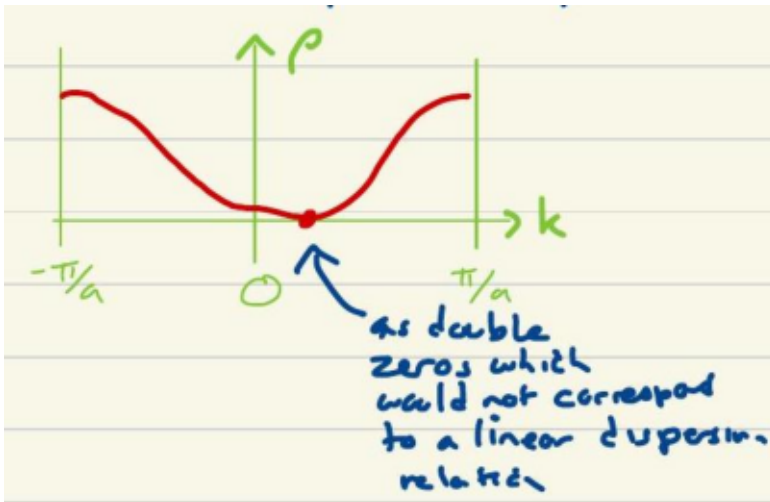


fig-redraw-017 (proposed)

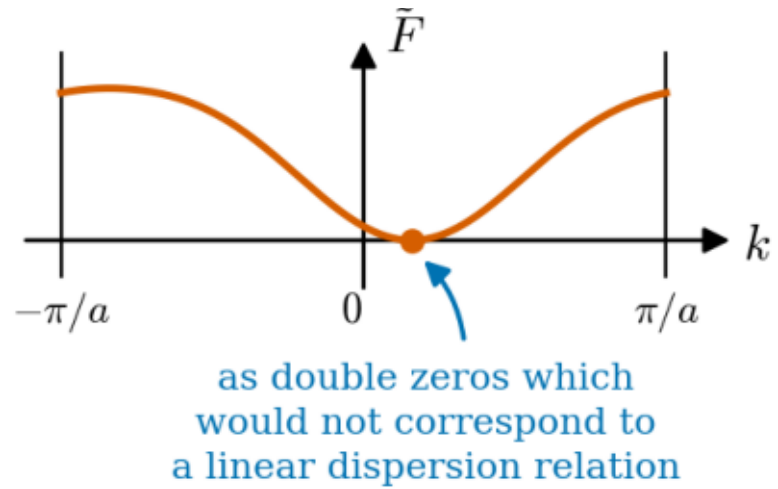


fig-notes-071.png (original)
 chapters/ch06-lattice-fermions/fermions.tex:472

$$C_{\mu\nu} = -\frac{i}{8} \left(\begin{array}{cc} \square \rightarrow & \square \rightarrow \\ \square \rightarrow & \square \rightarrow \end{array} - \begin{array}{cc} \square \rightarrow & \square \rightarrow \\ \square \rightarrow & \square \rightarrow \end{array} \right)$$

fig-redraw-071 (proposed)

$$C_{\mu\nu} = -\frac{i}{8} \left(\begin{array}{cc} \square \rightarrow & \square \rightarrow \\ \square \rightarrow & \square \rightarrow \end{array} - \begin{array}{cc} \square \rightarrow & \square \rightarrow \\ \square \rightarrow & \square \rightarrow \end{array} \right)$$

fig-notes-086.png (original)
 chapters/ch15-finite-volume/interactions.tex:94

$$\mathcal{A} = \text{diagram 1} + \text{diagram 2} + \text{diagram 3} + \dots$$

fig-redraw-086 (proposed)

$$\mathcal{A} = \text{diagram 1} + \text{diagram 2} + \text{diagram 3} + \dots$$

fig-redraw-046 (proposed)

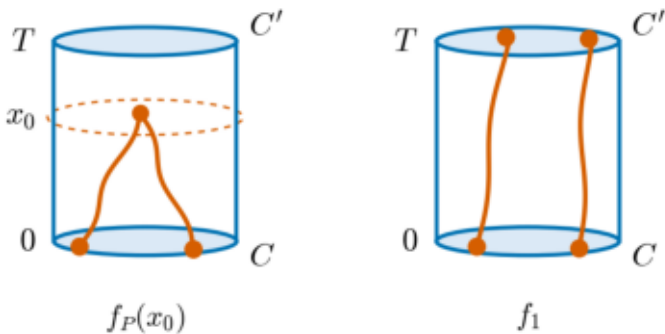
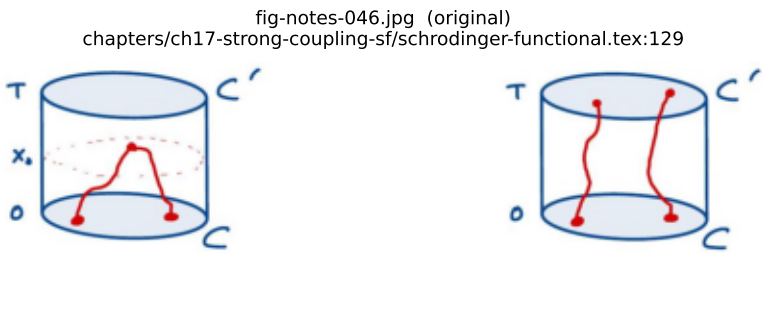


fig-notes-047.jpg (original)
chapters/ch17-strong-coupling-sf/schrodinger-functional.tex:162

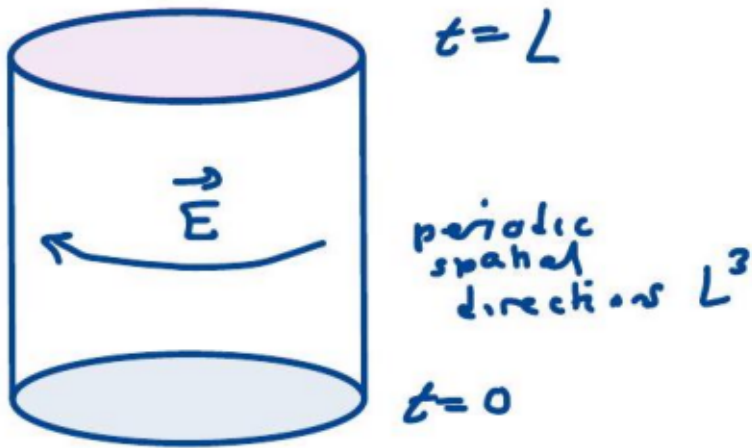


fig-redraw-047 (proposed)

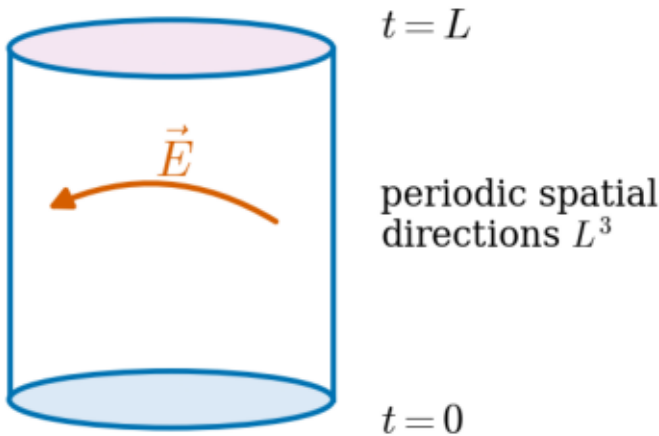


fig-redraw-048 (proposed)

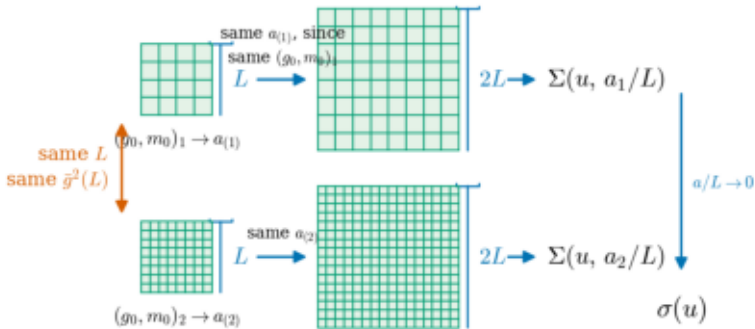
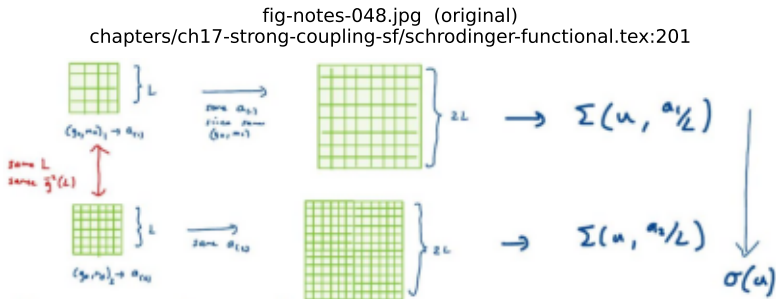


fig-notes-052.jpg (original)
chapters/ch19-chiral-gauge/chiral-gauge.tex:56

$$\mathcal{L} \supset \frac{1}{g^2} F_{\mu\nu} F_{\mu\nu} - \frac{i}{2} (\bar{\psi}_R \not{\partial} \psi_L + \text{h.c.}) + \underbrace{\tilde{\kappa} [(\partial_\mu A_\mu)^2 + (A_\mu^2)^2]}_{\tilde{\kappa} = \frac{1}{2g^2} \text{ gauge fixing}} + \underbrace{\rho A_\mu^2}_{\text{counterterms}} + \underbrace{\text{other counterterms + ghosts}}_{\text{needed if non-Abelian}}$$

fig-redraw-052 (proposed)

$$\mathcal{L} \supset \frac{1}{g^2} F_{\mu\nu} F_{\mu\nu} - \frac{i}{2} (\bar{\psi}_R \not{\partial} \psi_L + \text{h.c.}) + \underbrace{\tilde{\kappa} [(\partial_\mu A_\mu)^2 + (A_\mu^2)^2]}_{\tilde{\kappa} = \frac{1}{2g^2} \text{ gauge fixing}} + \underbrace{\rho A_\mu^2 + \text{other counterterms}}_{\text{counterterms}} + \underbrace{\text{ghosts}}_{\text{needed if non-Abelian}}$$

fig-notes-051.jpg (original)
chapters/ch19-chiral-gauge/chiral-gauge.tex:66

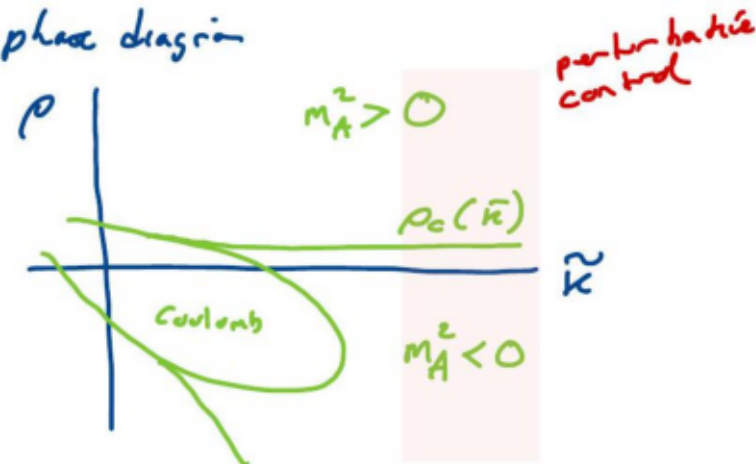


fig-redraw-051 (proposed)

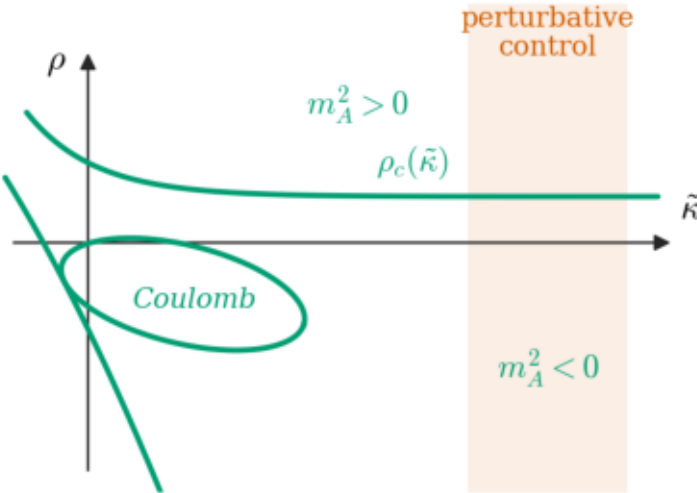


fig-notes-089.png (original)
chapters/ch20-lattice-susy/susy.tex:78

 $\sim a \int_{\text{BZ}} \frac{d^4 k}{k^3} \frac{k^2}{k} \sim a \Lambda^2 \sim \frac{\pi^2}{a} \bar{\psi} \psi$

fig-redraw-089 (proposed)

 $\sim a \int_{\text{BZ}} d^4 k \frac{k^2}{k^3} \frac{1}{k} \sim a \Lambda^2 \sim \frac{\pi^2}{a} \bar{\psi} \psi$

fig-notes-090.png (original)
 chapters/ch20-lattice-susy/susy.tex:180

$$\text{continuum} \quad \text{circle diagram} = 6g \int_{-\pi/a}^{\pi/a} \frac{dp}{2\pi} \frac{-ip+m}{p^2+m^2} = \frac{6g}{\pi} \arctan\left(\frac{\pi}{2ma}\right) \sim 3g + \mathcal{O}(ma)$$
 vanishes by $p \rightarrow -p$
 can cut off in other ways but a finite answer

$$\text{lattice} \quad \text{circle diagram} = \frac{6g}{L} \sum_{k=0}^{L-1} \frac{-2i \sin(\pi k/L) e^{i\pi k/L} + m_H a}{\sin^2(\pi k/L) + m_H^2 a^2} \xrightarrow{a \rightarrow 0} 6g$$
 where $m_H a = ma + 2 \sin^2\left(\frac{\pi k}{2L}\right)$ due to Wilson term

fig-redraw-090 (proposed)

$$\text{continuum} \quad \text{circle diagram} = 6g \int_{-\pi/a}^{\pi/a} \frac{dp}{2\pi} \frac{-ip+m}{p^2+m^2} = \frac{6g}{\pi} \arctan\left(\frac{\pi}{2ma}\right) \sim 3g + \mathcal{O}(ma)$$
 vanishes by $p \rightarrow -p$
 $\psi_2 = \psi_1$ here

$$\text{lattice} \quad \text{circle diagram} = \frac{6g}{L} \sum_{k=0}^{L-1} \frac{-2i \sin(\pi k/L) e^{i\pi k/L} + m_H a}{\sin^2(\pi k/L) + m_H^2 a^2} \xrightarrow{a \rightarrow 0} 6g$$
 can cut off in other ways, but is finite anyway

where $m_H a = ma + 2 \sin^2\left(\frac{\pi k}{2L}\right)$ due to Wilson term

fig-notes-087.png (original)
 chapters/ch21-perfect-actions/perfect-actions.tex:170

$$Q_\mu(n_B) = (1-6c) U_\mu(n) U_\mu(n+\hat{\mu}) + c \sum_{\nu \neq \mu} \left[\text{loop diagram with vertical line} \right] + \left[\text{loop diagram with horizontal line} \right]$$
 $\nu \neq \mu$

fig-redraw-087 (proposed)

$$Q_\mu(n_B) = (1-6c) U_\mu(n) U_\mu(n+\hat{\mu}) + c \sum_{\nu \neq \mu} \left[\text{loop diagram with vertical line} \right] + \left[\text{loop diagram with horizontal line} \right]$$
 $\nu \neq \mu$